MILITARY SPECIFICATION SHEET

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS); ROCKET CATAPULT MARK 16 MOD 1 (LOADED ASSEMBLY)

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the Mark 16 Mod 1 Rocket Catapult described herein shall consist of this specification and the latest issue of MIL-A-85097.

1. SCOPE

1.1 Scope. This specification sheet covers requirements unique to the Rocket Catapult Mk 16 Mod 1 (loaded assembly) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents are in addition to those listed in the cover specification, MIL-A-85097, and form a part of this specification sheet to the extent specified herein.

SPECIFICATIONS

Naval Air Systems Command (Code Ident 30003)

AS 3550	Propellant, N-43 (U)
AS 3554	Propellant, N-44 (U)

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

707AS129	Container Assembly, for Rocket Catapults
707AS130	Palletization Assembly, Rocket Catapults
736AS300	Assembly, Rocket Catapult, Mk 16 Mod 1
736AS190	Test Chamber
736AS191	Test Setup Assembly
736AS192	Adapter

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1377

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3. REQUIREMENTS

3.1 General. The requirements in this specification sheet unless in direct conflict with the basic specification shall be considered additional, not substitute. The Rocket Catapult shall conform to Drawing 736AS300.

3.2 Preproduction.

3.2.1 Propellant aging. When tested in accordance with MIL-A-85097, the cartridge and sustainer propellants shall meet the requirements of table I or table II, as applicable.

TABLE I. Propellant N-43 (AS 3550).

Property	Temp (°F)	Zero time		7-Year Equivalent			
				0 % Rel. Humidity		22 % Rel. Humidity	
		Min	Max	Min	Max	Min	Max
S_m (maximum stress) psi	77	157	256	157	269	157	276
E_m (elongation at S_m) %	77	18	28	9	28	16	28
Y_m (Young's modulus) psi	77	1014	1946	1014	3094	1014	3114
B_r (strand burn rate at 1000 psi) in/sec	77	0.60	0.67	0.60	0.71	0.60	0.71

TABLE II. Propellant N-44 (AS 3554).

Property	Temp (°F)	Zero time		7-Year Equivalent			
				0 % Rel. Humidity		22 % Rel. Humidity	
		Min	Max	Min	Max	Min	Max
S_m (maximum stress) psi	77	116	193	116	202	116	208
E_m (elongation at S_m) %	77	30	50	14	50	26	50
Y_m (Young's modulus) psi	77	589	1093	589	1737	589	1745
B_r (strand burn rate at 1000 psi) in/sec	77	0.55	0.67	0.60	0.71	0.60	0.71

3.2.2 Load. The rocket catapult shall withstand a static compression load of 18,000 pounds and a tension load of 8,700 pounds applied along the longitudinal axis at the attachment points without failure.

3.3 Preproduction/production.

3.3.1 Propellant. The propellant shall conform to the requirements of AS 3550 and AS 3554 for the following parameters at −65°F, 70°F, and 160°F: heat of explosion, burning rate, and physical properties.

3.3.2 Ballistic performance. The rocket catapult shall meet the requirements of table III when test-fired using an ejected weight of 315 pounds at 160°F and 410 pounds at −40°F and 70°F.

3.3.2.1 Output cartridge pressure. The head end output cartridge shall produce pressure values as specified in table III, when fired in a test chamber conforming to Drawing 736AS190, using test setup assembly and adapter in accordance with Drawings 736AS191 and 736AS192, respectively. The head end output cartridge shall reach maximum pressure within 10 milliseconds of rocket motor ignition as defined in MIL-A-85097.

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TABLE III. Ballistic performance requirements.

Parameter	−40° ± 5°F		70° ± 5°F		160° ± 5°F	
	Min	Max	Min	Max	Min	Max
Catapult ignition delay (ms)	0	5	0	5	0	5
Onset (g/s)	-	250[1]	-	250[1]	-	350[1]
Acceleration (g)	-	19[1]	-	19[1]	-	20[1]
Separation velocity[2] (fps)	41	-	-	-	-	-
Rocket ignition time (ms)	0	25	0	25	0	20
Rocket burn time (ms)	235	385[1]	200	285	175	245
Rocket action time (ms)	290	425[1]	240	355	210	275
Rocket motor resultant thrust (max)	3,165	4,935	3,700	5,800	4,750	7,300[1]
Rocket motor impulse (lb-s)	1,027[1]	-	1,027[1]	-	1,027[1]	-
Output cartridge (psi max)	1,080	2,100	-	-	1,080	2,100

[1] System requirements.
[2] At end of catapult stroke measured over the interval from 31 to 43 inches of travel.

3.3.2.2 Acceleration. During the first 0.015 second after initiation, any initial pulse shall not exceed 19 g's or exceed 5 g's for more than 0.010 second.

3.3.3 Rocket motor marking. The rocket motor serial number shall be stamped in black ink on the outer diameter of the rocket motor housing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Examinations and tests.

4.1.1 Preproduction.

4.1.1.1 Environmental.

4.1.1.1.1 Vibration. For the AEPS covered by this specification sheet, the following test procedures shall be used in lieu of those specified in the general specification. Two rocket catapults shall be subjected to vibration testing. One rocket catapult shall be temperature conditioned for a minimum of 8 hours at −40° ± 5°F and the other at 160° ± 5°F. The rocket catapults shall be maintained at specified conditioning temperature during these tests. If the rocket catapult must be removed from the conditioning chamber or if the rocket catapult temperature falls outside the specified temperature limits for more than 5 minutes, the testing shall stop and the rocket catapult shall be reconditioned. A reconditioning time of 2 minutes for every minute the rocket catapult spent outside the specified temperature envelope shall be required, up to a maximum of 8 hours. A continuous temperature history of the rocket catapult shall be kept for all conditioning and test time. Excitation shall be applied through the aircraft seat attachment fittings unless otherwise noted in the contract. Accelerometers to measure acceleration along the axis perperdicular to the longitudinal axis of the rocket catapult shall be placed midway between the attachment points. Longitudinal vibration shall be measured at either end of the rocket catapult.

4.1.1.1.1.1 Resonance survey. Resonance surveys shall be conducted on both rocket catapults in three axes at the maximum and minimum conditioning temperatures. Input vibratory acceleration of 5 g's shall be applied successively along each of the axes noted above when sweeping between 10 and 499 hertz. Input vibratory accelerations of 3 g's shall be applied for sweeps between 500 and 2,000 hertz. All resonances shall be as noted. During the resonance survey a permanent, continuous record of input vibratory acceleration and of unit response to the input vibratory acceleration shall be kept. This record shall be made on an instrument acceptable to the Government and shall include the response along each of the three axes. Frequency shall be varied at a rate not greater than 0.25 octave per minute.

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a. A resonance shall be defined as a frequency at which a clearly defined peak in the amplitude ratio (ratio of the output acceleration in any of the three igniter cartridge axes to the input acceleration as measured at the excitation attachment points) exceeds a value of 1.5 to 1.

b. If the resonance survey indicates a frequency band wherein the amplitude ratio exceeds 1.5 to 1 (as contrasted to a clearly defined peak) this entire band width shall be considered one resonant point. The minimum frequency band width which shall be considered a resonant band shall be 0.33 octave. If, also within this band width, there are clearly defined amplitude peaks, the most severe of these shall be considered an additional resonant point.

4.1.1.1.1.2 **Endurance vibration.** One rocket catapult shall be conditioned to $-40° \pm 5°F$ and one to $160° \pm 5°F$. Each rocket catapult shall be vibrated for 24 hours, 8 hours in each of the three axes. The vibration time in each axis shall be equally divided between the resonant frequencies noted in that axis during the resonant survey. A peak input acceleration of 5 g's shall be maintained at frequencies between 10 and 499 hertz and 3 g's for frequencies between 500 and 2,000 hertz. If no resonant frequency is detected in an axis, the rocket catapult shall be vibrated in that axis for a minimum of 2 hours at 5 g's input through the 10- to 499-hertz range and 6 hours at 3 g's input through the 500- to 2,000-hertz range. Total vibration time for the rocket catapults shall be 24 hours which shall include resonant and nonresonant frequencies, as applicable. At the conclusion of the tests, the rocket catapults shall be visually and radiographically inspected for damage without disassembly of the rocket catapult. The rocket catapults shall be fired at the same temperatures they were conditioned and vibrated.

4.2 **Preproduction/production.**

4.2.1 **Conditioning temperatures.** The conditioning temperatures and tolerances shall be as follows:

a. Cold: $-40° \pm 5°F$
b. Hot: $160° \pm 5°F$

4.2.2 **Conditioning period.** Time for conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: not less than 6 hours
b. Temperature cycling: not less than 6 hours

5. PACKAGING

5.1 **Packing.** Rocket catapults shall be packed as specified on Drawings 707AS129 and 707AS130.

6. NOTES

6.1 **Intended use.** The Rocket Ctapult Mk 16 Mod 1 is intended for use as a means of ejecting personnel from the A-4, A-7, and S-3A aircraft.

6.2 **Propellant description.** The following propellant information should be submitted with each production lot. The materials and components of each propellant batch should be listed with the manufacturer, lot number, and date of manufacture. The burning rates at 500, 1,000, and 1,500 psi performed at $-65°$, 70°, and 160°F for each pressure (three samples at each temperature and pressure per bucket) should also be included. In addition, include the results of tensile strength and elongation tests at $-65°$, 70°, and

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160°F with a pull rate of 2 inches/minute (three samples at each temperature per bucket), heat of explosion (3 samples per bucket), and verification that ingredients were properly weighed and added to each mix (mix sheet(s)).

6.4 Changes from previous issue. Revision letters are not used to denote changes due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-85097/1A (AS)	(AEPS): Rocket Catapult MK 16 MOD 1 (Loaded Assembly)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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